Report on *"The Large N Limit of Superconformal Field Theories and Supergravity"*

(J. Maldacena, hep-th/9711200)

Summary. The paper takes N coincident branes (D3, M2, M5, D1–D5, wrapped M5 on a four-cycle, and D3 branes making a four-dimensional extremal black hole) and applies a low-energy limit in which two things happen at once. The worldvolume theory decouples from the bulk and becomes a superconformal field theory. At the same time the supergravity solution, with energies held fixed, turns into its near-horizon throat, $AdS_{p+2} \times S^q$ (possibly times a compact manifold). The curvature of the throat in Planck units falls off as a power of $1/N$, so supergravity is reliable at large N. Because the field theory is unitary and contains the Hawking radiation of throat black holes, the paper argues that its large-N Hilbert space contains supergravity on AdS. It then conjectures that the full string/M-theory on these backgrounds is *exactly* dual to the corresponding CFT. Three facts support this: the superconformal group is the same as the AdS supersymmetry group (e.g. $SO(2,4) \times SO(6)$ bosonic part for $AdS_5 \times S^5$), the couplings map onto each other ($R^4/\alpha'^2 = 4\pi g N$), and the probe-brane action, to the extent checked, is fixed by broken superconformal symmetry.

1. Principal achievement

The main achievement is a concrete, well-motivated conjecture: a non-gravitational, unitary quantum field theory (four-dimensional $\mathcal{N} = 4$ $U(N)$ super-Yang–Mills) is equivalent to a quantum theory of gravity, type IIB string theory on $AdS_5 \times S^5$. The argument rests on a single, clean idea. One decoupling limit ($\alpha' \to 0$ with $U = r/\alpha'$ fixed) is read in two ways, once as the brane field theory and once as the near-horizon geometry. Several things follow from this:

- **Large-N gauge theory and strings become one problem.** The 't Hooft limit ($g \to 0$, gN fixed and large) gives free IIB strings on a space of radius $R^2/\alpha' = \sqrt{4\pi g N}$. This is an explicit realisation of 't Hooft's old expectation that large-N gauge theories are string theories, together with an identification of the large-N "master field" as a classical geometry.
- **A strong/weak duality with a computable side.** When perturbation theory in the gauge theory fails ($gN \gg 1$), classical supergravity is reliable, and $1/N$ corrections correspond to quantum-gravity effects such as Hawking radiation.
- **Earlier puzzles are explained.** The agreement of D3/D1–D5 absorption cross-sections and greybody factors with field theory is explained: the non-trivial physics sits in the AdS part of the geometry. The radial coordinate is interpreted as an energy scale (UV/IR), which gives a geometric picture of the renormalisation group.
- **A non-perturbative definition of quantum gravity.** Because the CFT can in principle be defined non-perturbatively, the conjecture gives a definition of string/M-theory on these backgrounds, including backgrounds with five non-compact dimensions.

The construction applies in the same way across several brane systems, which suggests a general principle rather than a coincidence of one example. This conceptual unification is the main contribution.

2. Principal drawback

The main drawback is that the conjecture is not operational: there is no dictionary, so the paper makes almost no non-trivial, checkable predictions. The author explicitly leaves the AdS boundary conditions ("and possibly also some boundary degrees of freedom") unspecified. There is also no rule for which CFT operator corresponds to which bulk field, or for how a CFT observable such as a correlation function is computed from the bulk. As a result:

- The evidence offered is mostly symmetry matching and quantities that are already protected. The author concedes that the probe Born–Infeld action is probably fixed entirely by superconformal symmetry and is "thus not a prediction". The earlier absorption agreements are known to follow from non-renormalisation theorems.
- The one dynamical comparison that is *not* protected, the near-extremal D3 entropy, disagrees with the free-field result by a numerical factor (3/4). This is not resolved; it is only attributed to unknown strong-coupling effects.
- The central step, that the CFT "includes" bulk supergravity, is a physical argument about limits (unitarity plus Hawking radiation into the throat), not a derivation. The leap to an *exact* duality at all N and g is a conjecture with no proposed test at strong coupling.
- The lower-supersymmetry cases are, as the author admits, sketchy. The (0,4) duality needs an unproven independence from vector moduli, and the $AdS_2 \times S^2$ case contains an unresolved puzzle (one obtains quantum mechanics with a large gap instead of a 1+1 dimensional CFT).

There are also minor presentational problems: the equation label `\fdbh` is used for two different equations, there are a few typos in the text and references, and the paper gives no explicit statement of what would falsify the proposal.

3. Most worthwhile follow-up

Proposal: build the field/operator dictionary and use it to make unprotected predictions. This directly addresses the main drawback, and every ingredient already exists, so the risk is low. Concretely:

1. **Spectrum matching.** Organise the known Kaluza–Klein spectrum of IIB supergravity on $AdS_5 \times S^5$ (Kim–Romans–van Nieuwenhuizen; Günaydin–Marcus, already cited in the paper) into $SU(2,2|4)$ multiplets. Compare these with large-N single-trace gauge-invariant operators, using the AdS relation $m^2R^2 = \Delta(\Delta-4)$ between mass and conformal dimension. The natural prediction is that the KK towers correspond to the chiral primaries $\mathrm{Tr}\, X^{(i_1} \cdots X^{i_k)}$ with $\Delta = k$. A sharper, *unprotected* prediction then follows. Massive string states have $m^2 \sim 1/\alpha'$, so every non-protected single-trace operator (e.g. Konishi, $\mathrm{Tr}\, X^i X^i$) must acquire an anomalous dimension growing like $\Delta \sim (gN)^{1/4}$ at strong coupling and drop out of the supergravity spectrum.
2. **A boundary-condition prescription.** Treat the boundary value of each bulk field as the source for the dual operator, and identify the on-shell bulk action, as a function of these boundary data, with the CFT generating functional. This fixes the boundary conditions the paper leaves open, and turns two- and three-point functions of chiral primaries, and thermodynamics at strong coupling, into classical supergravity calculations.
3. **Tests.** Compare the resulting three-point functions with free-field values; agreement would point to new non-renormalisation theorems. Treat the 3/4 entropy factor as the first strong-coupling prediction, $S(gN \to \infty)/S_{\text{free}} = 3/4$, rather than as an anomaly.

Why this, on balance. *Feasibility*: steps 1–2 need only classical supergravity in a maximally symmetric background and group theory, and the spectrum is already in the literature. *Impact*: the proposal becomes a calculational framework and a falsifiable one. It would produce strong-coupling predictions (the $(gN)^{1/4}$ growth of unprotected dimensions, specific correlators) that could later be checked independently. It would also be the foundation for every other application (black-hole thermodynamics, extensions to non-conformal and less supersymmetric theories). The more speculative directions raised in the paper, such as AdS_2, Euclidean CFT and de Sitter space, or twelve/thirteen-dimensional realisations, carry higher risk and depend on this groundwork anyway.

Remark (with hindsight). This is essentially the programme Gubser–Klebanov–Polyakov and Witten carried out in 1998. Its predictions, including the matching of protected three-point functions and the $\lambda^{1/4}$ growth of the Konishi dimension (later confirmed via integrability), were subsequently confirmed. This supports the judgement that it was the follow-up most worth pursuing.